Akshay Ganesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.88
Rao Junior College of Science	Class XII (HSC)	2021	97.83%
Podar International School	Class X (CBSE)	2019	97.20%

SKILLS

C | C++ | Python | SQL | CLI | AI | ML | RL | DL | DSA | OOP | DBMS | CV | Convex Optimisation | Exercise Science

Calculus | LA | Probability & Statistics | Discrete Mathematics | Olympiad Mathematics | Competitive Programming

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Foundations of Machine Learning [AI5000: FoML]. *IITH*

Head Placement Coordinator, Department of Artificial Intelligence. *IITH*

PROJECTS

'Pet Breed Classification: Vision Transformer vs. CNN' using PyTorch *IITH*

- Fine-tuned a pretrained Vision Transformer (ViT-B/16) in PyTorch to classify 37 cat and dog breeds (Oxford-IIIT Pets), achieving 93.7% test accuracy (macro-F1 0.936) and outperforming a ResNet-50 CNN baseline (92.1%) trained under identical settings. Analysed errors using confusion matrices and per-class metrics, and visualised the model's focus using attention rollout.

'Research-Paper Q&A Chatbot' using RAG *IITH*

- Built a Retrieval-Augmented Generation (RAG) chatbot that answers questions over 8 research papers using open-source models (BGE embeddings, Qwen2.5 LLM), with source-cited answers and a Gradio interface. Evaluated on a 30-question test set, reaching a 92% retrieval hit rate and 75% answer accuracy by tuning retrieval settings (top-k, chunk size), and analysed failure modes in answer generation and refusal behaviour.

'Car Purchase Prediction' using ML *TCET*

- Built a binary classification model for predicting prospective car purchasers based on the dataset obtained from a similar previous campaign. A total of 7 different ML classification algorithms were implemented, including an ensemble learning algorithm (Random Forest), and the trained model was able to achieve a high accuracy ratio of 93% in the test set with the non-linear RBF Kernel SVM classifier.

SCHOLASTIC ACHIEVEMENTS

Successful completion of the 'Deep Learning' NPTEL course conducted by IIT Madras (Course Instructor: Prof. Mitesh M. Khapra).

- Secured a consolidated score of 86 / 100 **[***Earned an Elite+Silver Certificate***]**.

Recognised as 'NPTEL Discipline Star' in the discipline of CSE by NPTEL.

- Completed a minimum of 60 weeks of courses in CSE (84 weeks in total) **[***Earned a Certificate of Appreciation***]**.

Successful completion of the 'Reinforcement Learning' NPTEL course conducted by IIT Madras (Course Instructor: Prof. Balaraman Ravindran).

- Secured a consolidated score of 69 / 100 **[***Earned an Elite Certificate***]**.

Secured AIR 1 in the 'Selected Topics in Algorithms' NPTEL course conducted by IIT Kharagpur (Course Instructor: Prof. Palash Dey).

- Secured a consolidated score of 93 / 100 **[***Earned an Elite+Gold Certificate***]**.

Successful completion of the 'Introduction to Soft Computing' NPTEL course conducted by IIT Kharagpur (Course Instructor: Prof. Debasis Samanta).

- Secured a consolidated score of 91 / 100 **[***Earned an Elite+Gold Certificate***]**.

Successful completion of the 'Python for Data Science' NPTEL course conducted by IIT Madras (Course Instructor: Prof. Ragunathan Rengasamy).

- Secured a consolidated score of 92 / 100 (Top 1% out of 11953 certified candidates) **[***Earned an Elite+Gold Certificate***]**.

Successful completion of the 'Introduction to Machine Learning' NPTEL course conducted by IIT Madras (Course Instructor: Prof. Balaraman Ravindran).

- Secured a consolidated score of 78 / 100 (Top 2% out of 1633 certified candidates) **[***Earned an Elite+Silver Certificate***]**.

Successful completion of the annual 'An Introduction to Artificial Intelligence' NPTEL course conducted by IIT Delhi (Course Instructor: Prof. Mausam).

- Secured a consolidated score of 54 / 100 **[***Pass Rate: 4.36% only***]**.

Successful completion of the 'Essential Mathematics for ML' NPTEL course conducted by IIT Roorkee (Course Instructors: Prof. S. Kumar and Prof. S. K. Gupta).

- Secured a consolidated score of 83 / 100 **[***Earned an Elite+Silver Certificate***]**.

Successful completion of the 'C Programming Language' course conducted under the Spoken Tutorial Project (FOSSEE) by IIT Bombay.

- Secured a consolidated score of 39 / 40 **[***Earned a Course Completion Certificate***]**.

Successful completion of the 'Discrete Mathematics' NPTEL course conducted by IIT Madras (Course Instructors: Prof. S. Iyengar and Prof. Prabuchandran K.J).

- Secured a consolidated score of 75 / 100 **[***Earned an Elite+Silver Certificate***]**.

Secured AIR 3 in the International Olympiad of Mathematics (iOM) exam

conducted by SilverZone Foundation.

- Earned a national-level bronze medal and a certificate of excellence.

RELEVANT COURSES

[AI5000]: Foundations of Machine Learning | **[AI5006]:** Linear Algebra-I | **[AI5106]:** Linear Algebra-II

[AI5030]: Probability and Stochastic Processes | **[AI5100]:** Deep Learning | **[AI5120]:** Topics in Optimization

[CS6450]: Visual Computing | **[CS6890]:** Fraud Analytics | **[CS6013]:** Advanced Data Structures and Algorithms

[LA5180]: Communication Skills: Advanced | **[AI5016]:** Industry Lecture Series

EXTRACURRICULAR

114 WPM: My best typing speed on a 60-second test (without punctuation) on Monkeytype.

National Service Scheme (NSS) Certificate: A 'Student Volunteer' in NSS (Unit B-34).